Image Segmentation with CNNs and Transformers

Applied Deep Learning — Chapter 21

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Foundation Books Project

Roadmap

From Boxes to Pixel Masks

Why Boxes Are Not Enough

Object detection returns a class, score, and four box coordinates per region.

A box around a polyp in an endoscopy image contains both polyp tissue and background tissue.

Applied questions the box cannot answer:

- How much tissue is involved?
- Is the boundary irregular?
- Are small protrusions systematically missed?

Segmentation keeps the location question but changes the output: one decision per pixel.

The Running Example: Kvasir-SEG

- Input: gastrointestinal endoscopy image.
- Target: binary mask — 1 inside the annotated polyp, 0 outside.
- Model first produces a probability per pixel; a threshold then converts probabilities into a final mask.

This is still deep learning for images, but the output is now spatially dense.

Treat it as a controlled applied example for learning segmentation workflow — not a clinical diagnostic system.

A Family of Tasks

Task	Output	Question answered
Classification	one label per image	does the image contain a polyp?
Detection	boxes, labels, scores	where is each polyp approximately?
Semantic seg.	one class label per pixel	which pixels are polyp pixels?
Instance seg.	one mask per object	which pixels belong to each lesion?
Panoptic seg.	full scene partition	road, sky, car 1, car 2, person 1?

The design question is not “newest architecture?” but “what output contract does the project need?”

Sparse vs Dense Outputs

Detection: a small list of numbers (sparse).

Segmentation at 384×384, binary:

$$H \times W = 384 \cdot 384 = 147{,}456 \text{ pixel decisions.}$$

- Harder to label, train, inspect, and evaluate.
- Can answer questions boxes cannot.
- Boundary pixels are the hard case — texture changes gradually.

Masks, Pixel Grids, and Metrics

A Mask Is a Set on a Pixel Grid

Pixel grid $\Omega = \{1, \ldots, H\} \times \{1, \ldots, W\}$.

Two equivalent views of a binary ground-truth mask:

- Function: $Y : \Omega \to \{0, 1\}$ (good for code).
- Set: $M = \{(i, j) \in \Omega : Y(i, j) = 1\}$ (good for metrics).

Probability map and threshold:

$$\widehat{M}_\tau = \{(i, j) \in \Omega : p_\theta(i, j) \geq \tau\}.$$

IoU and Dice

$$\mathrm{IoU}(M, \widehat{M}) = \frac{|M \cap \widehat{M}|}{|M \cup \widehat{M}|}, \qquad \mathrm{Dice}(M, \widehat{M}) = \frac{2|M \cap \widehat{M}|}{|M| + |\widehat{M}|}.$$

Both measure overlap; different normalizations.

Worked example. $|M| = 7$, $|\widehat{M}| = 8$, $|M \cap \widehat{M}| = 5$.

$$|M \cup \widehat{M}| = 7 + 8 - 5 = 10, \quad \mathrm{IoU} = \tfrac{5}{10} = 0.50, \quad \mathrm{Dice} = \tfrac{10}{15} \approx 0.667.$$

Dice especially common in medical segmentation, where foreground is small relative to background.

Why Pixel Accuracy Misleads

10×10 patch with two true polyp pixels.

All-background prediction: $98/100 = 0.98$ accuracy.

But:

- Predicted foreground set is empty.
- Foreground Dice $= 0$, foreground IoU $= 0$.
- Model has detected no polyp region at all.

Accuracy can hide misses when foreground pixels are rare.

Threshold Is Validation

Probabilities on a row of five pixels:

$$(0.20,\ 0.42,\ 0.55,\ 0.68,\ 0.91).$$

- $\tau = 0.30$: $(0, 1, 1, 1, 1)$
- $\tau = 0.50$: $(0, 0, 1, 1, 1)$
- $\tau = 0.70$: $(0, 0, 0, 0, 1)$

One trained model, three different masks.

Choose τ on validation data, then lock it before the final test evaluation.

Mean IoU and Boundary Distances

$$\mathrm{mIoU} = \frac{1}{K} \sum_{k=0}^{K-1} \mathrm{IoU}_k \, .$$

Document any rule for ignoring absent classes.

Hausdorff distance between mask boundaries ∂A, ∂B:

$$H(\partial A, \partial B) = \max\left\{ \sup_{a \in \partial A} \inf_{b \in \partial B} \|a - b\|_2, \; \sup_{b \in \partial B} \inf_{a \in \partial A} \|b - a\|_2 \right\} .$$

A good Dice score can still hide a clinically important boundary shift of a few pixels.

The Medical Segmentation Data Pipeline

Image-Mask Pairs

Every example is an ordered pair (X, Y):

- $X \in \mathbb{R}^{H \times W \times 3}$ — the endoscopy image.
- $Y : \Omega \to \{0, 1\}$ — binary mask of polyp pixels.

Common segmentation bugs are silent:

- image flipped but mask not flipped: model learns the wrong location;
- bilinear-resized class mask: yields invalid labels like 0.37;
- near-duplicate frames split across train/test: "score" becomes memorization.

Look before training: overlay images and masks before trusting any loss curve.

Paired vs Photometric Augmentation

Paired geometric same spatial transform on image and mask: flips, rotations, crops, resizing.

Photometric image only — brightness, contrast, color jitter. A class label is not a color measurement.

Interpolation rule:

- Natural images: bilinear OK (intermediate values meaningful).
- Class-index masks: nearest-neighbor (labels are discrete).

Splitting and Leakage

Write the split rule down before training.

- **Train:** updates parameters.
- **Validation:** selects checkpoints, thresholds, image sizes, learning rates.
- **Test:** reserved for the final reported number.

When patient, study, procedure, or video identifiers exist, split at that level.

Near-duplicate frames belong on the same side of train, validation, and test.

Pipelines vs Architectures

nnU-Net's lesson: a careful pipeline can be worth as much as a more fashionable model.

Sanity checklist before training:

1. Inspect ≥ 16 loaded examples.
2. Image and mask dimensions match after preprocessing.
3. Mask contains only expected values.
4. Augmentations move image and mask together.
5. Validation and test never used to fit weights or tune τ.

FCNs and U-Net

From Classifier to Dense Predictor

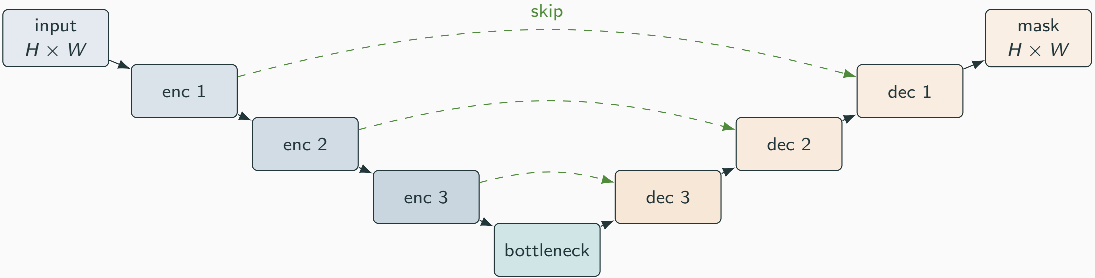
Classification networks compress an image into a feature vector.

Segmentation cannot end with one vector — it must preserve enough spatial information to predict on a grid.

Fully convolutional networks: replace dense classifier heads with convolutional prediction heads; upsample coarse feature maps back toward image resolution.

Tension: a 256×256 input downsampled $32\times$ becomes 8×8 — semantically rich, spatially too coarse for boundaries.

U-Net: Encoder + Decoder + Skips



Skip connections localize. Deep features decide what is present; shallow features place the boundary.

Pretrained Encoders Are Usually Worth It

A U-Net from random init can work given enough labeled data.

In a classroom-scale workflow, a pretrained encoder is usually a better start:

- Generic visual features come for free.
- Fine-tuning spends limited updates on the medical target.
- Same logic as classification transfer learning, applied to dense prediction.

Defaults considered later: U-Net (CNN baseline), SegFormer-B0 (modern Transformer baseline), nnU-Net (configuration-heavy biomedical).

Loss Functions and Training

Pixelwise Binary Cross-Entropy

For one pixel, target $y \in \{0,1\}$, predicted probability $p \in (0,1)$:

$$\mathrm{BCE}(y,p) = -y \log p - (1-y)\log(1-p).$$

Average over Ω:

$$\mathcal{L}_{\mathrm{BCE}}(\theta) = \frac{1}{|\Omega|}\sum_{(i,j)\in\Omega}\left[-Y \log p_\theta - (1-Y)\log(1-p_\theta)\right].$$

Sanity: $y=1, p=0.8 \Rightarrow -\log 0.8 \approx 0.223$; $y=1, p=0.01 \Rightarrow -\log 0.01 \approx 4.605$.

Confident misses on foreground are heavily penalized.

Soft Dice Loss

$$\mathrm{Dice}_{\mathrm{soft}} = \frac{2\sum Y\, p_\theta + \epsilon}{\sum Y + \sum p_\theta + \epsilon}, \qquad \mathcal{L}_{\mathrm{Dice}} = 1 - \mathrm{Dice}_{\mathrm{soft}}\,.$$

Combined loss:

$$\mathcal{L}(\theta) = \lambda\, \mathcal{L}_{\mathsf{BCE}} + (1-\lambda)\, \mathcal{L}_{\mathrm{Dice}}, \qquad 0 \le \lambda \le 1.$$

Three-pixel example. $Y = (1, 1, 0)$, $p = (0.8, 0.4, 0.3)$:

$$\mathrm{Dice}_{\mathrm{soft}} = \frac{2 \cdot 1.2}{2 + 1.5} = \frac{2.4}{3.5} \approx 0.686, \quad \mathcal{L}_{\mathrm{Dice}} \approx 0.314.$$

BCE scores pixels independently; Dice couples the whole foreground.

Speed and Accuracy Together

Reference run on a seeded Kvasir-SEG split (800/100/100):

Recipe	Size	Val Dice	Val IoU	Train time
Quick SegFormer-B0 baseline	256	0.8733	0.7752	49.4 s
Improved SegFormer-B0 recipe	384	0.8799	0.7855	99.9 s

Selected thresholds: 0.40 (256), 0.50 (384).

Validation selects between rows; the test set reports the chosen recipe once. The larger image gained 0.0065 Dice at roughly 2× cost — defensible, not automatic.

The Tiny-Overfit Diagnostic

Before training for many epochs:

- Train on 8–16 image-mask pairs.
- Check that the loss can be driven down.
- Inspect masks on those same examples.

Failure usually means a data, loss, shape, or learning-rate problem — *not* a model-capacity problem.

Fast feedback first. Tiny-overfit and low-resolution runs catch pipeline errors before expensive training.

Multi-Scale Context: DeepLab

Why Wider Context?

- Shiny endoscopy highlight: locally looks like a boundary; should not become its own foreground.
- Street-scene gray pixel: road, sidewalk, wall, or vehicle depending on surroundings.

Aggressive downsampling enlarges context but destroys boundary detail.

Need wider context at the chosen feature-map resolution.

Dilated Convolutions and ASPP

Dilated 3×3 kernel, dilation rate r:

$$y[i] = \sum_{m=0}^{k-1} w[m]\, x[i + r\, m].$$

Nine learned weights, 5×5 footprint at $r = 2$.

ASPP: parallel dilation rates so the decoder sees context at multiple scales without further downsampling.

DeepLabv3+ combines ASPP with an encoder-decoder structure for sharper boundaries.

Instance and Panoptic Segmentation

When Semantic Is Not Enough

- One visible polyp: binary semantic mask is enough.
- Many touching nuclei: a single foreground mask covers all nuclei pixels but cannot count them.

Counting, measuring, or tracking each object requires instance segmentation.

Mask R-CNN extends Faster R-CNN with a mask branch: the detector proposes regions; the branch predicts a mask aligned to each region.

Mask AP and Panoptic Segmentation

Mask Average Precision. Match predicted masks to ground-truth instance masks under IoU thresholds; compute AP over ranked predictions.

- Duplicate mask hurts precision.
- Missing object hurts recall.
- Poor overlap fails at higher IoU thresholds.

Panoptic segmentation. Every pixel gets an interpretation:

- "things": countable objects (cars, people).
- "stuff": amorphous regions (road, sky, tissue background).

Unifies semantic and instance under one task and quality measure.

Transformer-Based Segmentation

SegFormer: Simple and Efficient

- Hierarchical Transformer encoder $\Rightarrow$ features at multiple spatial scales.
- Lightweight MLP decoder fuses them into a semantic prediction.
- SegFormer-B0 small enough for fast classroom experiments while teaching a modern Transformer-based workflow.

Why hierarchies matter: fine features for boundaries, coarse features for object-level context. Kvasir-SEG needs both.

Mask Queries and Mask2Former

Modern segmenters predict a set of mask queries rather than a per-pixel classifier on a CNN feature map.

Schematic mask head: dot product between query and pixel embeddings,

$$\hat{m}_{q,p} = \sigma(q^\top e_p).$$

Training matches predicted masks to target masks; mask losses apply to matched pairs.

Mask2Former uses masked attention to focus computation, unifying semantic, instance, and panoptic segmentation under one mask-query view.

The Cost of Attention at High Resolution

Self-attention is expensive at high image resolution — and segmentation often needs high resolution.

Practical rules:

- Start with a small pretrained variant.
- Use a modest image size first.
- Fix the validation protocol before scaling.

Starting with the largest available model usually slows the feedback loop before the data pipeline is proven correct.

Promptable and Medical Foundation Models

SAM and MedSAM

Promptable segmentation: model accepts a prompt (point, box, coarse mask) and returns a mask for the indicated region.

- **Box prompt** less ambiguous than a single point in medical images.
- **SAM** trained on natural images.
- **MedSAM** adapts the idea to medical images, stronger across many medical tasks in that setting.

Promptable models help with annotation and bootstrapping — they do not remove the need for task-specific validation.

Choosing and Reporting a Segmentation Model

Picking a First Model

Situation	First choice	Main evidence
Small binary medical masks	SegFormer-B0 or U-Net	Dice, IoU, time, failures
Biomedical benchmark	nnU-Net	CV / held-out, config summary
Semantic scene seg.	DeepLabv3+ or SegFormer	mIoU, runtime, resolution
Separate instances	Mask R-CNN, Mask2Former	Mask AP, duplicates, misses
Interactive annotation	SAM or MedSAM	prompt type, edit time, validation

Validation-Selected Model

- Recipe A: validation Dice 0.78 at $\tau = 0.50$.
- Recipe B: validation Dice 0.81 at $\tau = 0.60$.

Select B with $\tau = 0.60$. If final test Dice is 0.79, report it.

Do not try new thresholds on the test set until the number improves.

Reference final test for the selected SegFormer-B0 recipe: Dice 0.8703, IoU 0.7705, ≈ 11.0 ms per test image.

What a Segmentation Report Must Include

Beyond the chosen number:

- Threshold and how it was selected.
- Image and mask preprocessing.
- Inference time at the reported resolution.
- Hardware identifier.
- Mask grid: clean success, false positive, false negative, boundary error, small target, hard image.

The mask is where the model stops being vague. If the overlay is wrong, the number has more explaining to do.

Read the Failure Grid for Patterns

- Thin missed rims $\Rightarrow$ threshold or boundary-loss work.
- Confident foreground outside the target $\Rightarrow$ check augmentation, leakage, negative examples.
- Vanishing small targets $\Rightarrow$ scale-aware sampling, loss reweighting.

Don't maximize one metric in isolation. A slightly higher Dice is less useful if the model is slow, memory-heavy, or weak on the targets that matter most.

Summary

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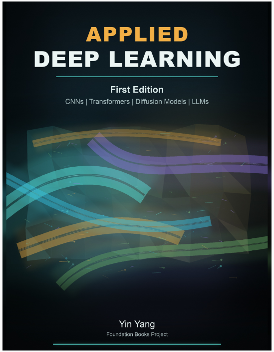
- **Output contract first:** pixel labels, instance masks, scene partition, or prompted masks.
- **Mask is a set on a pixel grid;** use IoU and Dice, not pixel accuracy alone.
- **Pipeline matters as much as architecture:** pair augmentations, resize masks safely, split by source.
- **Model families:** U-Net, DeepLabv3+, Mask R-CNN, SegFormer, Mask2Former, SAM/MedSAM.
- **Threshold is part of training discipline:** chosen on validation, reported once on test.
- **Show masks alongside numbers:** successes, boundary errors, and failures together.

Next: image generation — VAEs, GANs, and diffusion. The direction of attention reverses, but the evidence discipline does not.



Applied Deep Learning

Chapter overview slides for the book by Yin Yang.

Book page	`foundation-books.github.io/applied-deep-learning`
Code	`github.com/foundation-books/applied-deep-learning-code`
Print	`amazon.com/dp/B0GZF7QRSH`
Kindle	`amazon.com/dp/B0GZF9221X`
License	CC BY-NC-ND 4.0

 Full explanations, exercises, and companion-code context are in the book and linked repository.